



VARDHAMAN
COLLEGE OF ENGINEERING

VARDHAMAN COLLEGE OF ENGINEERING

(Autonomous)

Affiliated to **JNTUH**, Approved by **AICTE**, Accredited by **NAAC** with **A+ Grade**
Kacharam, Shamshabad, Hyderabad - 501 218, Telangana, India

LAB MANUAL

LAB DETAILS:

Laboratory:

Subject:

Student Name:

Roll Number:

Branch:

Academic Year:



Vardhaman College of Engineering

DEPARTMENT OF _____

Certificate

Certified that this is the bonafied record of practical work done by

Mr/Ms. KOLETI KEERTHI Roll Number: 25881A05FP

of B.Tech / M.Tech / MBA in the

A9513 - Python Programming Laboratory - 2025 - 2029 laboratory

during the year _____

No. of Experiments done: 21

Total No. of Experiments: _____

Date:

Staff Member / Incharge

HOD

Internal Examiner

External Examiner

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Mr./Ms. KOLETI KEERTHI

Roll Number: 25881A05FP

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Page No. 1	Week 1	Date: 2026-08-08
		Exp. No: 1

Experiment 1: Print Personal Details

Aim:

Write a program to print name, roll number, and branch. Inputs are read from system itself, follow the input order and output order.

Sample Input:

Asha
24CS101
CSE

Sample Output:

Asha
24CS101
CSE

Source Code:

```
#===== Write Your Logic Here =====
name=input()
rollNo=input()
branch=input()

print(name)
print(rollNo)
print(branch)
```

Result:

Status: COMPLETED

Test Cases Passed: 2 / 2

Score: 10.0

Total Submissions: 2

Page No. 2	Week 1	Date: 2026-08-02
		Exp. No: 2

Experiment 2: Display a String

Aim:

Write a program to take the string 'Welcome to Programming' as input and display it on the screen.

Sample Input:

Welcome to Programming

Sample Output:

Welcome to Programming

Source Code:

```
#===== Write Your Logic Here =====
print("Welcome to Programming")
```

Result:

Status: COMPLETED
Test Cases Passed: 1 / 1
Score: 10.0
Total Submissions: 1

Page No. 3	Week 1	Date: 2026-08-08
		Exp. No: 3

Experiment 3: Read and Display Mixed Values

Aim:

Write a program to read an integer, a floating-point value, and a string, then display all three values.

Sample Input:

```
7
3.5
Hello
```

Sample Output:

```
7
3.5
Hello
```

Source Code:

```
#===== Write Your Logic Here =====
integervalue=int(input())
decimalvalue=float(input())
text=input()

print(integervalue)
print(decimalvalue)
print(text)
```

Result:

Status: COMPLETED

Test Cases Passed: 14 / 14

Score: 10.0

Total Submissions: 1

Page No. 4	Week 2	Date: 2026-08-08
		Exp. No: 4

Experiment 4: Convert Fahrenheit to Centigrade

Aim:

Write a program to read a temperature in Fahrenheit and convert it to Centigrade using $C = (F - 32) * 5 / 9$.

Sample Input:

68

Sample Output:

20.00

Source Code:

```
#===== Write Your Logic Here =====
Fahrenheit=float(input())
c=(Fahrenheit-32)*5/9
print(f"{c:.2f}")
```

Result:

Status: COMPLETED

Test Cases Passed: 15 / 15

Score: 10.0

Total Submissions: 1

Page No. 5	Week 2	Date: 2026-08-08
		Exp. No: 5

Experiment 5: Calculate Area of a Triangle

Aim:

Write a program to read the base and height of a triangle and print its area.

Sample Input:

10

6

Sample Output:

30.0

Source Code:

```
#===== Write Your Logic Here =====
base=int(input())
height=int(input())

area=0.5*base*height
print(f"{area:.1f}")
```

Result:

Status: COMPLETED

Test Cases Passed: 14 / 14

Score: 10.0

Total Submissions: 1

Page No. 6	Week 2	Date: 2026-08-08
		Exp. No: 6

Experiment 6: Calculate Average Marks

Aim:

Write a program to read marks in five subjects and display their average.

Sample Input:

80
75
90
85
70

Sample Output:

80.0

Source Code:

```
#===== Write Your Logic Here =====
s1=int(input())
s2=int(input())
s3=int(input())
s4=int(input())
s5=int(input())

avg=(s1+s2+s3+s4+s5)/5
print(f"{avg:.1f}")
```

Result:

Status: COMPLETED

Test Cases Passed: 15 / 15

Score: 10.0

Total Submissions: 1

Page No. 7	Week 3	Date: 2026-08-10
		Exp. No: 7

Experiment 7: Demonstrate Comparison Operators

Aim:

Write a program that reads two numbers and displays the results of the comparison operators ==, !=, >, <, >=, and <=.

Sample Input:

4 7

Sample Output:

0 1 0 1 0 1

Source Code:

```
#===== Write Your Logic Here =====  
  
a, b =map(int,input().split())  
  
print(int(a==b),int(a!=b),int(a>b),int(a<b),int(a>=b),int(a<=b))
```

Result:

Status: COMPLETED

Test Cases Passed: 15 / 15

Score: 10.0

Total Submissions: 1

Page No. 8	Week 3	Date: 2026-08-10
		Exp. No: 8

Experiment 8: Swap Two Numbers

Aim:

Write a program to read two numbers and interchange their values.

Sample Input:

12 35

Sample Output:

35 12

Source Code:

```
#===== Write Your Logic Here =====  
a,b =input().split()  
print(b,a)
```

Result:

Status: COMPLETED

Test Cases Passed: 15 / 15

Score: 10.0

Total Submissions: 1

Page No.	Week 3	Date: 2026-08-26
9		Exp. No: 9

Experiment 9: Demonstrate Operators

Aim:

Write a program to demonstrate the following operators using the given input values.

Unary Operator: Print the unary negation of the integer n (i.e., -n).

Bitwise Left Shift: Print the result of $x \ll s$.

Logical Operator: Print 1 if both logical values a and b are true (non-zero); otherwise print 0.

Membership Operator: Check whether the integer k is present in the list [1, 2, 3]. Print 1 if present; otherwise print 0.

Identity Operator: Create a list items = [1, 2, 3] and compare it with itself using the identity operator (or equivalent reference comparison in your programming language). Print 1 if the comparison is true; otherwise print 0.

Print each result on a new line in the above order.

Sample Input:

```
5
1 3
1 1
2
```

Sample Output:

```
-5
8
1
1
1
```

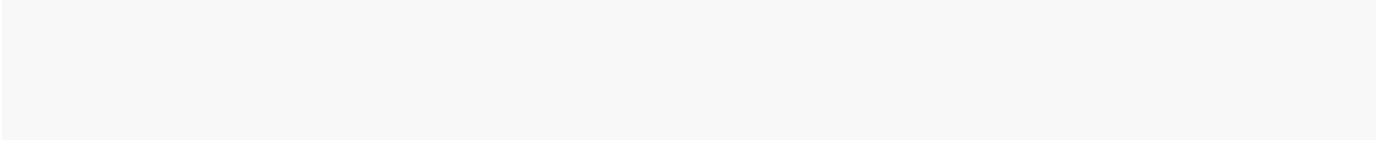
Source Code:

```
##### Write Your Logic Here #####
n=int(input())
x,s=map(int,input().split())
a,b=map(int,input().split())
k=int(input())

print(-n)
print(x<<s)
print(1 if a and b else 0)
print(1 if k in [1,2,3] else 0)

item =[1,2,3]
print(1 if item is item else 0)
```

Page No. 10	Week 3	Date: 2026-08-26
		Exp. No: 9



Result:

Status: COMPLETED

Test Cases Passed: 11 / 11

Score: 10.0

Total Submissions: 1

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Page No. 11	Week 4	Date: 2026-08-12
		Exp. No: 10

Experiment 10: Check Even or Odd Number

Aim:

Given a number, write a program to determine whether it is even or odd.

Sample Input:

18

Sample Output:

Even

Source Code:

```
##### Declare Imports here if required #####

##### Declare Global Variables / Functions here if required #####

n = int(input())
##### Declare Local Variables / Functions here if required #####

##### Write Your Logic Here #####

if n % 2==0:
    print("Even")
else:
    print("Odd")

pass
```

Result:

Status: COMPLETED

Test Cases Passed: 15 / 15

Score: 10.0

Total Submissions: 1

Page No. 12	Week 4	Date: 2026-08-26
		Exp. No: 11

Experiment 11: Find the Greatest of Three Numbers

Aim:

Write a program to read three integers and display the greatest value.

Sample Input:

18 42 31

Sample Output:

42

Source Code:

```
##### Write Your Logic Here #####  
  
a,b,c=map(int,input().split())  
print(max(a,b,c))
```

Result:

Status: COMPLETED

Test Cases Passed: 14 / 14

Score: 10.0

Total Submissions: 1

Page No. 13	Week 4	Date: 2026-08-26
		Exp. No: 12

Experiment 12: Demonstrate Nested If

Aim:

Write a program using nested if statements to determine whether a number is positive, negative, or zero and, when positive, whether it is even or odd.

Sample Input:

14

Sample Output:

Positive Even

Source Code:

```
##### Write Your Logic Here #####
n=int(input())

if n > 0:
    if n % 2 == 0:
        print("Positive Even")
    else:
        print("Positive Odd")
else:
    if n<0:
        print("Negative")
    else:
        print("Zero")
```

Result:

Status: COMPLETED

Test Cases Passed: 15 / 15

Score: 10.0

Total Submissions: 1

Page No. 14	Week 5	Date: 2026-08-26
		Exp. No: 13

Experiment 13: Reverse a Number

Aim:

Given an integer N, reverse its digits.

Sample Input:

12345

Sample Output:

54321

Source Code:

```
##### Declare Imports here if required #####

##### Declare Global Variables / Functions here if required #####

##### Declare Local Variables / Functions here if required #####

##### Write Your Logic Here #####
n = int(input())

rev = 0

while n > 0:
    digit= n % 10
    rev = rev * 10 + digit
    n = n // 10

print(rev)
```

Result:

Status: COMPLETED

Test Cases Passed: 15 / 15

Score: 10.0

Total Submissions: 1

Page No. 15	Week 5	Date: 2026-08-26
		Exp. No: 14

Experiment 14: Factorial of N

Aim:

Given an integer N, find N! (factorial of N).

Sample Input:

5

Sample Output:

120

Source Code:

```
##### Declare Imports here if required #####

n = int(input())
##### Write Your Logic Here #####

fact = 1

for i in range(1, n+1):
    fact = fact * i
print(fact)
```

Result:

Status: COMPLETED

Test Cases Passed: 15 / 15

Score: 10.0

Total Submissions: 1

Page No. 16	Week 5	Date: 2026-08-26
		Exp. No: 15

Experiment 15: Find All Factors of a Number

Aim:

Print all factors of N sorted in ascending order.

Sample Input:

12

Sample Output:

1 2 3 4 6 12

Source Code:

```
##### Declare Imports here if required #####

##### Declare Global Variables / Functions here if required #####

##### Declare Local Variables / Functions here if required #####

n = int(input())

##### Write Your Logic Here #####
factors = []

for i in range(1, n+1):
    if n % i == 0:
        factors.append(i)

print(*factors)
```

Result:

Status: COMPLETED

Test Cases Passed: 15 / 15

Score: 10.0

Total Submissions: 1

Page No. 17	Week 5	Date: 2026-08-27
		Exp. No: 16

Experiment 16: Fibonacci Series

Aim:

Given an integer N, print the first N numbers of the Fibonacci series (0, 1, 1, 2, 3...).

Sample Input:

7

Sample Output:

0 1 1 2 3 5 8

Source Code:

```
##### Declare Imports here if required #####

##### Declare Global Variables / Functions here if required #####

n = int(input())

##### Declare Local Variables / Functions here if required #####

##### Write Your Logic Here #####
a,b = 0, 1
x =[]

for i in range(n):
    x.append(str(a))

    a, b = b, a + b
print(" ".join(x))
```

Result:

Status: COMPLETED

Test Cases Passed: 14 / 14

Score: 10.0

Total Submissions: 1

Page No. 18	Week 5	Date: 2026-08-29
		Exp. No: 17

Experiment 17: List Prime Numbers up to N

Aim:

Write a program that lists all prime numbers from 1 to N. A prime number is greater than 1 and divisible only by 1 and itself.

Sample Input:

10

Sample Output:

2
3
5
7

Source Code:

```
#===== Declare Imports here if required =====

#===== Declare Global Variables / Functions here if required =====

#===== Declare Local Variables / Functions here if required =====

#===== Write Your Logic Here =====

n = int(input())

for i in range(2, n + 1):
    count = 0

    for j in range(1 ,i + 1):
        if i % j == 0:
            count += 1

    if count == 2:
        print(i)

if n == 1:
    print("NULL")
```

Page No. 19	Week 5	Date: 2026-08-29
		Exp. No: 17



Result:

Status: COMPLETED

Test Cases Passed: 15 / 15

Score: 10.0

Total Submissions: 4

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Page No.	Week 6	Date: 2026-08-29
20		Exp. No: 18

Experiment 18: String Analysis and Transformations

Aim:

Write a program that reads a string and displays its length; the string repeated 10 times; its first character; first and last three characters; reverse; seventh character or 'Not exist'; the string without its first and last characters; uppercase form; and a version with every character replaced by a space.

Sample Input:

programming

Sample Output:

11
programmingprogrammingprogrammingprogrammingprogrammingprogrammingprogrammingpro
p
pro
ing
gnimmargorp
m
rogrammin
PROGRAMMING

Source Code:

```
##### Write Your Logic Here #####
s = input()

print(len(s))
print(s * 10)
print(s[0])
print(s[:3])
print(s[-3:])
print(s[::-1])

if len(s) >= 7:
    print(s[6])
else:
    print("Not exist")

print(s[1:-1])
print(s.upper())
print(" " * len(s))
```


Page No. 21	Week 6	Date: 2026-08-29 Exp. No: 18
Result: Status: COMPLETED Test Cases Passed: 7 / 7 Score: 10.0 Total Submissions: 1 25881A05FP		

Page No. 22	Week 6	Date: 2026-08-29
		Exp. No: 19

Experiment 19: Demonstrate String Concatenation, Repetition, and Membership

Aim:

Write a program that reads two strings, a substring, and an integer. Concatenate the two strings, repeat the first string the given number of times, and determine whether the given substring exists in the first string.

Sample Input:

```
Hello
World
ell
3
```

Sample Output:

```
HelloWorld
HelloHelloHello
1
```

Source Code:

```
##### Write Your Logic Here #####
str1 = input()
str2 = input()
substring = input()
n = int(input())

print(str1 + str2)
print(str1 * n)

if substring in str1:
    print(1)
else:
    print(0)
```

Page No. 23	Week 6	Date: 2026-08-29 Exp. No: 19
Result: Status: COMPLETED Test Cases Passed: 5 / 5 Score: 10.0 Total Submissions: 1 25881A05FP		

Page No. 24	Week 6	Date: 2026-08-29
		Exp. No: 20

Experiment 20: Demonstrate Built in String Methods

Aim:

Write a program that reads a string and demonstrates the upper(), lower(), strip(), and find() string methods.

Sample Input:

```
hello
ll
```

Sample Output:

```
HELLO
hello
hello
2
```

Source Code:

```
#===== Write Your Logic Here =====
text = input()
searchString = input()

print(text.upper())
print(text.lower())
print(text.strip())

index = text.strip().find(searchString)
print(index)
```

Result:

Status: COMPLETED

Test Cases Passed: 5 / 5

Score: 10.0

Total Submissions: 1

Page No. 25	Week 6	Date: 2026-09-01
		Exp. No: 21

Experiment 21: Calculate Vowel Frequency

Aim:

Write a program to calculate the frequency of vowels in a given text string. A vowel is any of the letters: a, e, i, o, u. The calculation must be case-insensitive, meaning 'A' and 'a' should both be counted as vowels. The input text may contain spaces and other characters.

Sample Input:

Artificial Intelligence

Sample Output:

10

Source Code:

```
##### Declare Imports here if required #####

##### Declare Global Variables / Functions here if required #####
s=input()
sl=s.lower()
count=0
for i in sl:
    if i in ["a","e","i","o","u"]:
        count += 1

print(count)
```

Result:

Status: COMPLETED

Test Cases Passed: 15 / 15

Score: 10.0

Total Submissions: 2



VARDHAMAN
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LAB MANUAL

Vardhaman College of Engineering

Kacharam, Shamshabad, Hyderabad - 501 218, Telangana, India

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